

Data Science & Machine Learning Math Practice Workbook

Linear Algebra (50 Questions)

1. Compute the norm of vector $(3, -4, 12)$.
2. Find the unit vector in direction of $(1,2,2)$.
3. Compute dot product of $(1,2,3)$ and $(4,-1,2)$.
4. Find angle between $(1,0,1)$ and $(2,1,3)$.
5. Check if $(1,2,3)$ and $(2,4,6)$ are linearly independent.
6. Solve system: $x+y=5$, $2x-y=1$.
7. Multiply matrices $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$.
8. Find determinant of $\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$.
9. Find inverse of $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$.
10. Find rank of $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$.
11. Reduce matrix to row echelon form.
12. Check if matrix is symmetric.
13. Compute trace of matrix.
14. Find eigenvalues of $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.
15. Find eigenvectors of $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.
16. Diagonalize matrix $\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$.
17. Check if matrix is diagonalizable.
18. Compute projection of vector.
19. Find basis of span.
20. Check subspace conditions.
21. Compute covariance matrix of dataset.
22. Show covariance matrix is symmetric.

23. Compute SVD of simple matrix.
24. Explain PCA via eigen decomposition.
25. Find null space of matrix.
26. Find column space.
27. Compute determinant of 3x3 matrix.
28. Use cofactor expansion.
29. Solve system using inverse.
30. Check orthogonality.
31. Compute Gram-Schmidt process.
32. Find orthonormal basis.
33. Compute matrix norm.
34. Find spectral norm.
35. Compute Frobenius norm.
36. Find condition number.
37. Explain curse of dimensionality.
38. Matrix factorization example.
39. Compute eigen decomposition.
40. Explain variance maximization.
41. Compute PCA for small dataset.
42. Find singular values.
43. Check positive definiteness.
44. Compute quadratic form.
45. Matrix multiplication properties.
46. Transpose properties proof.
47. Rank-nullity theorem example.
48. Linear transformation matrix.

49. Kernel of transformation.

50. Image of transformation.

Probability (50 Questions)

1. Find probability of even number on die.
2. Find probability of at least one head in 2 tosses.
3. Compute $P(A \cup B)$ given probabilities.
4. Find conditional probability $P(A|B)$.
5. Check independence of events.
6. Define random variable.
7. Find expectation of discrete variable.
8. Find variance.
9. Compute binomial probability.
10. Compute Poisson probability.
11. Find normal probability using Z.
12. Convert to standard normal.
13. Compute CDF from PDF.
14. Find PDF normalization constant.
15. Compute expectation from PDF.
16. Find variance of uniform distribution.
17. Compute covariance.
18. Check independence vs uncorrelated.
19. Use Bayes theorem problem.
20. Disease testing problem.
21. Spam classification example.
22. Compute entropy.
23. Compute joint probability.

24. Find marginal distribution.
25. Conditional expectation.
26. Law of total probability.
27. Random variable transformation.
28. Jacobian simple example.
29. Moment generating function.
30. Find mean using MGF.
31. Variance using MGF.
32. Continuous distribution problems.
33. Discrete distribution problems.
34. Bernoulli distribution.
35. Geometric distribution.
36. Exponential distribution.
37. Memoryless property.
38. Central moments.
39. Skewness calculation.
40. Kurtosis intuition.
41. Mixture distributions.
42. Likelihood function.
43. Posterior probability.
44. Prior vs posterior.
45. MLE estimation.
46. MAP estimation.
47. Sampling distribution.
48. Monte Carlo idea.
49. Law of large numbers.

50. Variance reduction.

Statistics (50 Questions)

1. Compute mean of dataset.
2. Find median and mode.
3. Compute variance.
4. Compute standard deviation.
5. Identify outliers.
6. Z-score calculation.
7. Normalize data.
8. Standardize data.
9. Sampling methods.
10. Bias in sampling.
11. Central limit theorem example.
12. Confidence interval calculation.
13. 95% CI for mean.
14. Hypothesis testing example.
15. Z-test problem.
16. t-test problem.
17. p-value interpretation.
18. Type I error example.
19. Type II error example.
20. Power of test.
21. Linear regression fit.
22. Least squares derivation.
23. Compute regression coefficients.
24. Residual analysis.

25. R-squared calculation.
26. Correlation coefficient.
27. Covariance interpretation.
28. ANOVA basics.
29. Chi-square test.
30. Goodness of fit.
31. Logistic regression idea.
32. Odds ratio.
33. Likelihood estimation.
34. Overfitting example.
35. Underfitting example.
36. Bias-variance tradeoff.
37. Cross-validation.
38. K-fold CV.
39. Regularization L1.
40. Regularization L2.
41. Ridge regression.
42. Lasso regression.
43. Feature scaling impact.
44. Model evaluation metrics.
45. Precision recall.
46. F1 score.
47. ROC curve idea.
48. AUC interpretation.
49. Bootstrap sampling.
50. Jackknife method.

Differential Calculus (50 Questions)

1. Find derivative of x^2 .
2. Find derivative of $\sin(x)$.
3. Find derivative of e^x .
4. Compute limit $\sin x/x$.
5. Check continuity.
6. Chain rule example.
7. Differentiate $(x^2+1)^5$.
8. Implicit differentiation.
9. Log differentiation.
10. Partial derivative of x^2+y^2 .
11. Gradient of function.
12. Directional derivative.
13. Hessian matrix.
14. Critical points.
15. Second derivative test.
16. Convex function check.
17. Optimize quadratic function.
18. Find maxima/minima.
19. Lagrange multipliers.
20. Constraint optimization.
21. Multivariable chain rule.
22. Jacobian matrix.
23. Taylor expansion.
24. Approximation using Taylor.
25. Gradient descent step.

26. Compute gradient of loss.
27. Learning rate effect.
28. Convergence criteria.
29. Stochastic gradient descent.
30. Mini-batch GD.
31. Momentum update.
32. Adam optimizer idea.
33. Vanishing gradient.
34. Exploding gradient.
35. Backpropagation basics.
36. Loss function derivative.
37. Mean squared error gradient.
38. Log loss gradient.
39. Sigmoid derivative.
40. ReLU derivative.
41. Softmax derivative.
42. Cross entropy loss.
43. Convex vs non-convex.
44. Saddle points.
45. Optimization landscape.
46. Regularization gradient.
47. Early stopping idea.
48. Gradient clipping.
49. Numerical differentiation.
50. Finite difference method.